

1 CHEMISTRY 1021 LABORATORY SYLLABUS

LAB SCHEDULE Fall 2024

Monday **OR** Tuesday
SCI 3132

<u>DATE</u>	<u>EXPERIMENT</u>
Aug 26 - 27	Laboratory Safety (p. v to viii)
Sept 2 - 3	Labor Day - No Lab
Sept 9 - 10	Exp 2: Molecular Models of Organic Compounds
Sept 16 - 17	Exp 3: Properties of Hydrocarbons
Sept 23 - 24	Exp 4: Properties of <i>Cis-Trans</i> Isomers
Sept 30 – Oct 1	Exp 5: Reactions of Alcohols
Oct 7 - 8	Exp 6: Reactions of Aldehydes and Ketones
Oct 14 - 15	Fall Break - No Lab
Oct 21 - 22	Mid-Term Exam
Oct 28 - 29	Exp 7: Extraction of Organic Acids and Bases
Nov 4 - 5	Exp 9: Fatty Acids, Esters, and the Preparation of Soap
Nov 11 - 12	Exp 11: Analysis of Carbohydrates, Fats, and Proteins in Snack Foods
Nov 18 - 19	Exp 10: Titration of Vitamin C
Nov 25 - 26	Exp 12: Optimum Conditions for an Enzyme – Lactase
Dec 2 - 3	Final Exam

YOU are responsible for keeping up with which experiment comes next. Before lab, read the experiment and try to understand the basic ideas. There will be a quiz at the beginning of the lab period covering the fundamental ideas (not details like how many mL of liquid...) of each experiment.

For “Laboratory Safety” you must show up for the Safety Briefing, take the Safety Quiz, and complete and turn in page xi/xii and the Safety Agreement (ix). Handout copies of all the pages will be provided to you for this week only. No one will be permitted to start work without completing the Safety Orientation.

REQUIRED - You must purchase and bring to lab a *new* laboratory manual. It is *Introductory Chemistry 2 Laboratory Manual*, Chemistry 1021, Tammy Melton, Hayden-McNeil, 2023; ISBN 978-1-5339-6418-2. The lab manual is available on campus and in area bookstores.

REQUIRED - You must purchase and bring to lab approved safety goggles. You must *wear* the goggles for all the experiments (beginning with Exp 3 on Sept 16-18).

SAFETY RULES

1. You must purchase **AND WEAR APPROVED SAFETY GOGGLES** while in lab. They are available in the bookstore. Do not buy goggles from Wal-Mart or Target. They are not approved for laboratory work. Read the statement on laboratory safety in the laboratory manual. Additional safety information will be discussed at the first lab meeting.
2. You must dress appropriately. No shorts, no ripped jeans with exposed skin, or open-weave yoga pants. No open-toed shoes: flip-flops, sandals, mules. Full sneakers are the best choice. Shirts must cover the belly and upper body. No tank tops, spaghetti straps, bare backs, bare shoulders, or ponchos with fringe. Wear old clothes; they may get ruined.
3. NO FOOD OR DRINK IN LAB. Bring it in and it will be taken from you and thrown out. Put water bottles or drink containers INSIDE the backpack BEFORE stepping inside the room.

OTHER RULES

1. Labs start **on time**. Arrive early. During the first five minutes a quiz will be given on the day's lab. This means that you must READ the experiment before coming to lab. It also means that if you are late, you will not be permitted to take the quiz.
2. **If you arrive to lab AFTER the pre-lab briefing begins, you will not be permitted to stay and perform the experiment. This is for safety reasons; you may have missed vital safety information. THIS RULE WILL BE ENFORCED–DON'T BE LATE!**
3. Bring your lab manual, safety goggles, and pencil to each experiment. It is also a good idea to bring your textbook. A cubby (NOT FLOOR) area will be designated for you to put your backpack. Do not keep it at your station. It becomes a tripping hazard or can cause a spill or accident at your workstation.
4. Labs count 10 points each. (2 pts for quiz + 8 pts for lab report.) We drop the lowest grade; **if you miss a lab, that's the grade which will be dropped. There are no make-up labs!** We do 11 experiments – lowest = 10 exp @ 10 pts = 100 pts. We will do two lab exams @ 50 pts each = 100 pts. Total lab points = 200. Your lecture instructor will explain how the lab grade is calculated into your overall final grade for the course.
5. **YOU MUST PASS THE LAB TO PASS THE COURSE.** This means if you miss too many experiments or do poorly on the lab exams, you will get an F regardless of your lecture average.
6. There is a mid-term lab exam over the first half of experiments, and a lab final over the second half. You will need a calculator for the lab final.

HONESTY

1. Lab partners consist of TWO (2) people. Don't work in larger groups of 3 or more unless you are specifically told to do so. If you have trouble finding a partner, one will be assigned to you.

Sneaky groups working in 3s, 4s, etc. without specifically being told to do so will be penalized.

2. During the quiz, you are expected to have your lab book closed, no notes out, and to keep your eyes on your own quiz paper. Do not talk or wander about the room. Wait until the quiz is over before you open your drawer to begin finding equipment. Cheating violations will be dealt with most severely and can include an F in the course, suspension, or expulsion from the University.

3. The use of, or copying from old lab reports, current or old, either yours or another's, for an experiment is cheating. You must do your own original work at the scheduled time of the experiment.

4. Do not copy the work of others in the room. Do your own work! You may collaborate with your lab partner on questions, but the work and the wording on the answers must be your own.